

MIDC Grand Challenge 2026

Challenge Details

Problem Statements (choose one):

1. **Smart Pothole Detection & Repair** — A ready-to-implement solution to detect potholes early (sensor-based, image-based, or crowdsourced) and enable fast, durable repair with a mechanism to prevent recurrence.
2. **Carbon Capture, Emission Management & Trading** — Solutions that help industries capture, treat or reduce emissions and improve ambient air quality, with a reliable way to calculate and report emission rates — including solutions that enable participation in carbon trading through verified emission reductions.
3. **Wastewater, Effluent Treatment, Desalination & Waste Recycling** — Ready-to-implement water effluent treatment solutions for CETPs and individual units, along with water reuse, desalination and advanced water-recovery technologies, paired with efficient waste collection, segregation and recycling systems.
4. **Renewable & Sustainable Green Energy Solutions** — Solar, wind, biogas, waste-to-energy or other renewable technologies deployable at estate or unit level, commercially viable at scale.
5. **AI-Driven Solutions for Industrial Infrastructure** — AI-based solutions for any aspect of industrial estate management — land/asset monitoring, predictive maintenance, resource optimisation, or investor/citizen service delivery — with a clear, measurable impact.

1. Problem Statement No. *

☒ 1

☐ 2

☐ 3

☐ 4

☐ 5

2. Problem Statement Title: *

PotholeMukt MIDC : AI Pothole Detector & Repair Guidance

3. Sector/ Theme *

Pothole Detection & Repair

4. Sub-Sector: *

AI-Driven Road Inspection & Patching

5. Location/ MIDC Industrial Area Proposed for Pilot: *

Taloja MIDC

6. Type of Applications *

- ☒ Innovation/ Idea Stage
- ☐ Prototype/ MVP
- ☐ Proof of Concept Stage (PoC)
- ☐ Existing Solution Ready for Deployment
- ☐ Scale-up of existing Solution

Applicant Details

7. Applicant category *

- ☒ Individual Innovator
- ☐ Student/Student Team
- ☐ Startup

- ☐ MSME
- ☐ Company
- ☐ LLP/ Partnerships
- ☐ Academic Institutions
- ☐ R&D Institutions
- ☐ Incubator/ Accelerator
- ☐ NGO/ Section 8 Company
- ☐ Consortium

Organisation Details

8. Name of Applicant/ Organisation *

Ritesh Singh and Ishant Wankhede

9. Legal Status *

Indian

10. Year of establishment/ Incorporation *

NA

11. DPIIT Startup Recognition No. (If Application. otherwise put NA) *

NA

12. Udyam Registration No. (If Applicable) *

0

13. PAN No. *

0

14. GSTIN (If Applicable) *

0

15. Registered Address *

2501, Bhoomi Elite, Sector 28, Nerul Navi Mumbai, 400706

16. Link to your Website *

<https://ishantwankhede.github.io/midc-hackathon/>

17. Organisation Profile *

NA

18. Registered Full Address with PIN Code *

2501, Bhoomi Elite, Sector 28, Nerul Navi Mumbai, 400706

19. City *

Navi Mumbai

20. District *

Navi Mumbai

21. State *

Maharashtra

22. Email Address *

ishant30oct@gmail.com

Lead Applicant / SPOC

23. Full Name of the SPOC *

Ishant Wankhede

24. Designation *

Senior Data Scientist

25. Email Address *

ishant30oct@gmail.com

26. Phone Number *

8011025342

27. Linkedin Profile URL *

<https://www.linkedin.com/in/ishantwankhede>

Team Details

28. Core Team Details (Name, Designation/Role, Qualification, Experience, Role in Challenge) *

Ritesh Singh, Senior Construction Program Manager, MS in Civil Engineering, 10years, Technical Program Manager

Team Capability (Briefly Describe)

29. Technical expertise *

Construction Civil Engineering and Artificial Intelligence

30. Domain expertise *

Construction Civil Engineering and Artificial Intelligence

31. Product development capability *

Full Stack

32. Previous relevant experience *

10years

33. Experience in implementing projects/pilots *

Combined 20 years for multiple projects across construction, healthcare and retail

Understanding of the Problem Statement

34. What is the Problem? *

Industrial estates like MIDC carry continuous 24/7 heavy container traffic (40-tonne trucks). Combined with heavy Maharashtra monsoons, road surfaces degrade rapidly into severe potholes.

Currently, road inspection relies on slow, manual reporting, meaning potholes go unaddressed for weeks, leading to expensive vehicle breakdowns, accidents, and logistics delays.

Furthermore, conventional road repairs require daytime lane closures and standard patches wash away quickly during rains, causing pothole recurrence and wasted maintenance funds.

There is a lack of real-time automated detection, traffic-smart repair scheduling, and long-lasting patching solutions.

35. Who is affected? *

Potholes in MIDC estates impact four primary groups: logistics & truck operators who suffer vehicle damage and delivery delays; factory owners facing supply chain disruptions and higher freight costs; MIDC authorities struggling with manual inspection and recurring maintenance expenses; and thousands of factory workers and daily commuters exposed to traffic jams and road safety hazards every day.

36. Current method/process *

Currently, road inspection relies on slow, manual reporting, meaning potholes go unaddressed for weeks, leading to expensive vehicle breakdowns, accidents, and logistics delays.

Furthermore, conventional road repairs require daytime lane closures and standard patches wash away quickly during rains, causing pothole recurrence and wasted maintenance funds.

37. Key Limitations *

1. Reactive & Manual Inspection: Current monitoring relies on physical complaints or manual supervisor visits, leaving 70%+ of industrial roads uninspected for weeks.
2. Weather Sensitivity: Standard hot-mix asphalt cannot be laid during heavy rains/monsoons, halting repairs for months.
3. Lack of Traffic Awareness: Maintenance is often scheduled during peak daytime hours, causing severe traffic jams for heavy trucks and commuters.
4. No Recurrence Tracking: There is no system to track whether a patched pothole reopens after 3 months, leading to repeated contractor payments for the same spot.

38. Scale of the problem in MIDC *

The scale of the problem spans over 289 MIDC industrial estates in Maharashtra containing 10,000+ km of road network. Heavy industrial hubs handle hundreds of thousands of 40-tonne cargo trucks daily under heavy monsoon rains (2,000mm+ annually). This results in thousands of potholes forming every season, causing millions of rupees in recurring repair costs, 20–30%

higher vehicle maintenance for logistics operators, and frequent traffic gridlocks across major industrial supply chains.

39. What Solutions Currently Exist? *

smartphone accelerometer and 3D laser inspection vans

40. Why are they inadequate? *

while smartphone accelerometer apps exist, they suffer high false-positive rates and cannot estimate repair costs, and 3D laser inspection vans are far too expensive for continuous industrial estate monitoring.

41. What gap does your solution address? *

PotholeMukt MIDC addresses the gap between slow manual inspections and long-lasting road maintenance. It bridges the gap in real-time visibility using Vision-AI fleet scanning across industrial roads; eliminates cost ambiguity by auto-calculating repair material and timeline estimates; prevents industrial traffic jams through smart day vs. night dispatching based on traffic density; and solves pothole recurrence by introducing monsoon-resistant polymer patching tracked via digital quality audits.

42. Solution Summary (Describe your proposed solution in simple, non-technical language.) *

PotholeMukt MIDC solves industrial road problems in 3 simple steps:

1. Automated Spotting: Small cameras on regular moving city buses and garbage trucks scan industrial roads daily, pinning every pothole on an interactive digital map.
2. Smart Estimations & Scheduling: For every pothole, the system automatically calculates repair costs, required materials, and recommends fixing it at night or low-traffic hours so heavy container trucks don't get stuck in traffic jams.
3. Fast & Permanent Repairs: Repair crews use a rapid-curing, rain-proof material that dries in just 30 minutes, ensuring heavy trucks won't break the patch even during monsoon rains.

43. What is innovative or substantially better about your solution? *

What makes PotholeMukt MIDC substantially better is its end-to-end intelligence and easy deployment. Instead of buying expensive specialized vehicles, we mount smart cameras on existing municipal fleets (garbage trucks/buses) for daily road scanning. Beyond simple

detection, our system automatically calculates exact repair costs, rough material volume, and tentative timelines. It protects industrial logistics by scheduling repairs day vs. night based on traffic density.

44. Technology Used *

- ☒ AI/ ML
- ☐ IoT
- ☐ Robotics/ Automation
- ☐ GIS
- ☐ Digital Twin
- ☐ Drone Technology
- ☐ Clean Technology
- ☐ Renewable energy
- ☐ Advanced Materials
- ☐ Data Analytics
- ☐ Software/ SaaS
- ☐ Hardware
- ☐ Others

45. If "Others " please specify the details *

Drone Technology

46. Technology Readiness Level *

- ☐ Idea
- ☒ Concept
- ☐ Prototype
- ☐ MVP
- ☐ PoC
- ☐ Pilot Deployed
- ☐ Commercially Deployed

47. Current Status, Describe the present stage of development and major milestones already achieved. *

Currently we are in Research & Development, building concept and gathering subject matter expertise.

MIDC Pilot Proposal

48. Proposed Pilot Location (MIDC Area, District, Proposed Site, Appromixate Area/ Infrastructure required) *

- MIDC Area: Taloja MIDC
- District: Raigad District (for Taloja)
- Proposed Site: Taloja: Taloja MIDC Main Industrial Spine Road & Chemical Belt (Sectors 1 to 24)
- Approximate Area / Stretch: 5 km to 10 km stretch of heavy industrial freight roads

49. **Pilot Plan** Describe how the solution will be implemented in the selected MIDC area. Include:

- Activities
- Infrastructure required
- Manpower

- Technology requirements
- Implementation methodology

*

- [1. SCAN] → Fleet cameras capture road video daily during regular routes.
- [2. ESTIMATE] → AI auto-calculates pothole area, depth, material mix, and repair cost.
- [3. SCHEDULE] → System schedules patching during low-traffic night hours (11 PM–4 AM).
- [4. PATCH] → Technicians apply rain-proof polymer asphalt (traffic resumes in 30 mins).
- [5. AUDIT] → GIS dashboard tracks patched location for 365 days to verify non-recurrence.

50. Implementation Timeline

Milestone: Outcome	Duration	Expected
1. Development/ Modification		
2. Installation		
3. Testing		
4. Pilot		
5. Performance Assessment		

*

- Milestone 1: Development / Modification
 - Duration: 2 Weeks (Weeks 1–2)
 - Expected Outcome: Refined Vision-AI defect detection model, GIS map integration for Taloja MIDC, and automated repair cost calculation logic.
- Milestone 2: Installation
 - Duration: 1 Week (Week 3)
 - Expected Outcome: Mounting 3x camera pods on pilot vehicles (patrol/buses); setting up material storage at Taloja MIDC office.
- Milestone 3: Testing
 - Duration: 2 Weeks (Weeks 4–5)
 - Expected Outcome: Processing 100+ km of road feeds, verifying AI accuracy (>90%), and calibrating cost/timeline estimates.
- Milestone 4: Pilot Execution
 - Duration: 4 Weeks (Weeks 6–9)
 - Expected Outcome: Live daily road scanning, automated work-ticket generation, and night-time repair demo on 1-km industrial stretch using rapid polymer asphalt.
- Milestone 5: Performance Assessment
 - Duration: 3 Weeks (Weeks 10–12)

- Expected Outcome: 30-day durability tracking of patched roads under heavy 40-tonne truck loads, and final pilot report submission to MIDC.

51. Support Required from MIDC *

- ☐ Land/Site
- ☐ Electricity
- ☐ Water Utilities
- ☒ Data
- ☐ Connectivity
- ☐ Access to Industrial Units
- ☐ Permission/ Approvals
- ☐ Technical Support
- ☐ Financial Support
- ☐ Other:

52. If "Others" provide the details *

NA



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